01.05 · THEORY + PAST-PAPER INDEX

Nuclear Physics

The atomic nucleus, binding energy, radioactivity and stellar reactions

START HERE

Nuclear physics explains how the nucleus is assembled, why reactions release energy and how unstable nuclei decay. The same ideas power stars, produce neutrinos and set the composition of the universe. This lecture begins with nuclear notation and mass–energy, then develops binding, decay and stellar fusion.

LEARNING GOALS

✓	Read proton, neutron and nucleon numbers from a nuclide symbol.
✓	Relate mass defect to nuclear binding energy and reaction energy.
✓	Use the exponential decay law, half-life and activity consistently.
✓	Describe fusion, fission and the role of neutrinos in stellar physics.

1 · The nucleus and the language of nuclides

An atom consists of a nucleus and electrons. The nucleus contains protons, with positive electric charge, and neutrons, which have no electric charge. The atomic number Z is the number of protons and identifies the element. The neutron number is N; the mass number A counts all nucleons, so A = Z + N.

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Nuclide label: mass number A, proton number Z · N = A − Z
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Nuclides of one element with different neutron numbers are isotopes. The atomic mass in a periodic table is usually a weighted average over naturally occurring isotopes, not the mass of one nucleus. For nuclear calculations, use masses from a consistent convention and track whether the quoted value includes electrons.

FORCES INSIDE THE NUCLEUS

The strong nuclear interaction binds nearby nucleons and overcomes the electric repulsion between protons at short distances. The strong force is attractive over a limited range and becomes repulsive at very short separation. Nuclear radius grows approximately as $R \approx r_0 A^{(1/3)}$, with r_0 about 1.2 femtometres.

2 · Mass defect and binding energy

A bound nucleus has less mass than the same protons and neutrons separated far apart. The missing mass is the mass defect. The energy equivalent of that mass is the binding energy: it is the energy that would have to be supplied to separate the nucleus completely into its constituent nucleons.

$$\Delta m = Zm_p + Nm_n - m_{nucleus} \cdot E_{bind} = \Delta m\, c^2$$

The conversion factor 1 atomic mass unit × c^2 is approximately 931.5 MeV. Binding energy is commonly quoted per nucleon, E_{bind}/A, to compare nuclei of different sizes. A larger binding energy per nucleon generally means a more tightly bound nucleus, although stability also depends on proton-to-neutron balance.

MASS CONVENTIONS MATTER

If neutral atomic masses are used, the electron masses must be handled consistently. In many reaction equations the electron numbers cancel on both sides, making atomic masses convenient. Never subtract a nuclear mass from an atomic mass without checking what each number includes.

The binding-energy-per-nucleon curve rises rapidly for light nuclei, peaks near iron and nickel, and declines slowly for very heavy nuclei. This pattern explains why combining light nuclei (fusion) and splitting very heavy nuclei (fission) can release energy.

3 · Nuclear reactions, fusion and fission

A nuclear reaction must conserve electric charge, total nucleon number and energy–momentum. It also obeys additional particle-number rules, such as lepton-number conservation in weak interactions. Write every reactant and product explicitly before comparing masses.

$$Q = (m_{initial} - m_{final})c^2$$

A positive Q means the final products have less rest mass and the difference can emerge as kinetic energy, radiation or energy carried by neutrinos. A negative Q reaction requires energy input. One atomic mass unit corresponds to about 931.5 MeV; 1 MeV = 1.602×10^{-13} J.

Fusion joins light nuclei. In stars, the net conversion of four protons into a helium-4 nucleus releases energy through a chain of reactions; positrons and electron neutrinos are also produced. Fission splits a heavy nucleus into smaller fragments, often releasing neutrons and energy. The detailed reaction path depends on temperature, density and composition.

WHY FUSION CAN RELEASE ENERGY

The helium nucleus is more tightly bound per nucleon than hydrogen. The final bound products therefore have less total rest mass than the starting particles. The mass difference is not destroyed; it appears as released energy according to $E = mc^2$.

4 · Radioactivity and the decay clock

An unstable nucleus can transform spontaneously into a more stable configuration. Radioactive decay is probabilistic for an individual nucleus, but a large sample follows a predictable exponential law. The decay constant λ is the probability per unit time for a nucleus to decay.

$$N(t) = N_0 e^{(-\lambda t)} \cdot t_{1/2} = \ln 2 / \lambda$$

The half-life $t_{1/2}$ is the time for half of the undecayed nuclei to remain. After n half-lives, the remaining fraction is (1/2)^n. Half-life does not depend on how much of the sample you start with or on ordinary changes in temperature and pressure.

$$\text{Activity: } A = \lambda N \cdot 1 \text{ becquerel} = 1 \text{ decay } s^{-1}$$

Activity is the rate of decays in the sample; it decreases with time along with N. To estimate an age, compare a measured parent-to-daughter ratio or remaining parent fraction with a known decay law, while accounting for any daughter material present at formation and any later gains or losses.

COMMON RADIOACTIVE EMISSIONS

Alpha decay emits a helium-4 nucleus; beta-minus decay converts a neutron into a proton, electron and antineutrino; beta-plus decay converts a proton into a neutron, positron and neutrino; gamma emission releases a high-energy photon as a nucleus loses excitation energy. Conservation laws determine the balanced reaction equation.

5 · Nuclear energy in stars

A star's core is hot and dense enough for some nuclei to approach closely despite electric repulsion. Quantum tunnelling allows fusion even when the typical thermal energy is below the classical Coulomb barrier. The proton–proton chain is the main hydrogen-fusion pathway in Sun-like stars; the carbon–nitrogen–oxygen cycle is important in hotter, more massive stars.

Fusion supplies energy that helps support a star against gravity. The core must be hot enough for reactions, but energy does not leave the surface immediately: it is transported outward through radiation and convection. The rate depends strongly on temperature and composition, so stellar fusion is a regulated process rather than a simple burn like a chemical flame.

NEUTRINOS: DIRECT MESSENGERS FROM THE CORE

Photons scatter many times inside the Sun and can take a very long time to diffuse outward. Neutrinos interact only weakly with matter, so most escape almost directly from the reaction site. Detecting solar neutrinos therefore tests nuclear reactions in the core. The same weak interaction makes neutrinos difficult to detect: enormous detectors and long observing times are needed.

When a massive star exhausts its available nuclear fuel, the balance between pressure and gravity can fail. Collapse and explosive events create conditions for additional nucleosynthesis and produce neutrinos. The observed neutrino signal, elemental abundances and remnant compact objects each probe a different stage of that history.

6 · A dependable nuclear calculation

Balance the reaction first, including neutrinos, electrons or positrons. Count protons and nucleons on each side. Then choose a consistent mass table, find the mass difference, convert to energy and check the sign. For radioactive samples, convert the time and half-life to the same units before evaluating the exponential.

$$1\ u = 1.6605 \times 10^{-27}\ kg \cdot 1\ u\ c^2 \approx 931.5\ MeV$$

For astrophysical scale estimates, distinguish energy per reaction from total power. Multiply the reaction rate by the energy released per reaction, and state whether energy carried by neutrinos is included in the observable luminosity. Report whether a quantity is per nucleus, per mole or per kilogram.

WHAT THE 'Q' TAG MEANS

Some nuclear topics are marked qualitative in the IOAA syllabus. That means deep quantitative proficiency is not required by the syllabus; the physical ideas and conservation reasoning still matter. Follow the information and constants supplied by the particular problem.

PAST IOAA PROBLEM REFERENCES

The following questions are from official past IOAA problem sets.

Use each year, round and question label to locate the original problem in its official paper.

01	IOAA 2008 · TH S4
02	IOAA 2009 · TH P10
03	IOAA 2009 · TH P11
04	IOAA 2010 · TH S5
05	IOAA 2011 · TH S14
06	IOAA 2012 · TH S6
07	IOAA 2013 · TH S3
08	IOAA 2014 · TH P3
09	IOAA 2014 · TH P6
10	IOAA 2015 · TH S12
11	IOAA 2017 · TH T6
12	IOAA 2018 · TH T11
13	IOAA 2023 · TH Q8
14	IOAA 2024 · TH T4
15	IOAA 2025 · TH T07
16	IOAA 2025 · TH T10